

State management & data layer (Riverpod)

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Master technical specification — **Volume 4 (Clients)**, nano-detail deep-dive. This document **deepens** the *Riverpod data layer* (§8) and the *FFI surface* (§3) of the pass-1 client overview [04_UI §4/§5, doc 03 §3/§8] into an implementation-ready specification of the helix_domain state layer: the Riverpod provider graph, the rule that **the UI is a pure function of the Rust core's status stream** (CI2), the flutter_rust_bridge StreamSink → Stream mechanics, the helix_api REST + WS/SSE client, how **Helix Console** folds control-plane events into Riverpod state, and the offline / optimistic (intent-vs-actual) honesty machine. SPEC-ONLY: it describes **what to build**, not the shipping product.

Boundary with sibling docs. This document **consumes**: the FFI surface + TunnelStatus mirror owned by doc 03 [03 §3]; the orchestrator tokio::broadcast<TunnelStatus> stream + state machine owned by v02-data-plane/orchestrator-and-state.md [02-ORCH §4/§5]; the Transport contract [02-TT]; and the control-plane REST/WS/SSE + event taxonomy owned by doc 02. It **owns** everything from the Dart HelixCore wrapper upward: how providers turn streams into reactive UI, how controllers issue commands, how events are reduced, and the offline honesty rule. It does **NOT** own the Rust orchestrator internals, the per-platform TunnelPlatform shim implementations (doc 03 §4–5, doc 06), the helix_design widgets (doc 03 §7), or the REST/WS wire schema (doc 02).

Evidence base. Citations inline by id: [04_UI] = 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md; [04_ARCH §N] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md; [03 §N] = final/03-client-core-and-ui.md; [02-ORCH §N] = final/v02-data-plane/orchestrator-and-state.md; [02-TT §N] = final/v02-data-plane/transport-trait.md; [SYN §N] = v09-research/_SYNTHESIS.md; [research-flutter_fffi] / [research-ios_android] = the cited platform research digests. Any claim not grounded in the evidence base is tagged UNVERIFIED per constitution §11.4.6 — never fabricated.

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0. Position, ownership, and invariants

0.1 What this document owns

The state layer lives in the `helix_domain` melos package [03 §2] (`packages/helix_domain/lib/`). It owns five contracts:

#	Contract	Owned here	Consumed from
S1	The Dart TunnelStatus projection + the mapping from the orchestrator wire enum to the FFI/Dart enum	§2	wire enum [02-ORCH §4.1]; FFI mirror [03 §3.1]
S2	The provider graph — every Riverpod provider, its dependencies, its lifetime, its override seams for flavor + test	§4	—
S3	tunnelStatusProvider + ConnectController — the stream→state binding and the command controller	§5	HelixCore FFI [03 §3.2], TunnelPlatform [03 §4]
S4	The helix_api consumption layer — REST repositories + the WS/SSE client + the Console event reducer	§6, §7	REST/WS schema, event taxonomy → doc 02
S5	The offline / optimistic honesty machine — intent vs actual, never green-on-intent (CI2)	§8	—

0.2 What this document does NOT own

- The Rust orchestrator (its loops, state machine, kill-switch, broadcast *sender*) — owned by [02-ORCH]; this doc owns the *receiver* side.
- The `helix-ffi` Rust source `api.rs` (the hand-authored FFI surface) — owned by [03 §3.1]; this doc consumes its **generated** Dart mirror.
- The per-platform `TunnelPlatform` shim impls (Swift/Kotlin/C#/C++/ArkTS) — owned by [03 §4–5]; this doc owns only the Dart `TunnelPlatform` *abstract contract* it drives and the provider that injects the per-OS impl.
- REST/WS/SSE wire schema, auth, the `device.revoked/route.changed/...` event payloads — owned by doc 02; this doc owns only the Dart *reducer* over them.
- Widgets / tokens / theming — owned by [03 §7].

0.3 Invariants this document inherits and tightens

#	Invariant	Source	State-layer tightening
CI2	The UI is a pure function of the core's status stream. No polling.	[03 §0.1 CI2, 04_UI §4.2]	§1, §5: the ONLY writer of the rendered connection state is <code>tunnelStatusProvider</code> , fed by the FFI <code>StreamSink</code> . A widget never derives "connected" from its own command success.
CI5	State must be announced, not just colored.	[03 §0.1 CI5]	§2: the Dart enum carries a <code>semanticLabel</code> projection so ally readers announce state from the same source the color reads.
H-I1	Intent ≠ actual. The user's wish to be connected is local optimistic state; <i>actual</i> protection is only the core stream.	[04_UI §4.4, 03 §8.4]	§8: a dedicated <code>connectIntentProvider</code> , never conflated with <code>tunnelStatusProvider</code> .
H-I2	A Danger state overrides any intent, paints red immediately, and cannot be optimistically suppressed.	[03 §8.4]	§8.3.
H-I3	Console links <code>helix_api</code> ONLY (no <code>helix_core_ffi</code>) — so it builds to Web.	[03 §0.1 CI3]	§4.4: <code>helixCoreProvider</code> is overridden to <code>NoCore()</code> in the Console flavor; <code>tunnelStatusProvider</code> is never watched in Console screens.
H-I4	Every provider is overridable for flavor wiring (§4.4) and for tests (fake <code>HelixCore</code> + fake API with the <i>same generated model types</i>).	[03 §12, 04_UI §12]	§4.5, §11.

#	Invariant	Source	State-layer tightening
H-15	Lag is not an error. A broadcast/WS receiver that falls behind re-reads latest, never surfaces an error to the user.	[02-ORCH §4.6]	§3.4, §10.

1. The single load-bearing rule: UI = pure function of the status stream

A VPN client's one safety-critical claim is *"am I protected right now?"* The architecture makes that claim un-faleable by construction: the rendered connection state is **derived only** from a stream the Rust core pushes [04_UI §4.2, 03 §0.1 C12]. The button reflects the stream; it never reflects its own onTap.

flowchart LR

```

    CORE["helix-core orchestrator (Rust)<br/>
>tokio::broadcast<&TunnelStatus_wire><br/>[02-ORCH §4]"]
    FFI["helix-ffi adapter (Rust)<br/>project wire→FFI enum
(§2.3)<br/>StreamSink<&TunnelStatus>"]
    FRB["frb_generated.dart<br/>StreamSink → Dart Stream (§3)"]
    SP["tunnelStatusProvider<br/>
>StreamProvider<&TunnelStatus> (§5.1)"]
    W["ConnectButton / StatusChip / ShieldIndicator<br/>
>ref.watch(tunnelStatusProvider)"]
    CTRL["ConnectController (§5.2)<br/>start/stop COMMANDS only"]
    SHIM["TunnelPlatform shim (§5.3)"]

    CORE --> FFI --> FRB --> SP --> W
    W -. "onTap (a command, NOT a state write)" .-> CTRL
    CTRL -. "startTunnel / stopTunnel" .-> SHIM
    SHIM -. "creates TUN, links core" .-> CORE
    classDef truth fill:#10B98122,stroke:#10B981;
    class CORE,FFI,FRB,SP,W truth;

```

The asymmetry is the whole point [03 §3.3]: **commands flow down** (widget → controller → shim → core), **truth flows up** (core → FFI → provider → widget), and the two paths never short-circuit. A connect() that succeeds does not let the UI paint green — only the subsequent Connected event on the stream does. This structurally eliminates the classic "UI says connected while the OS tunnel is down" defect (it is the UI-layer mirror of [02-ORCH §13.1] "single state authority" O-I10).

2. The TunnelStatus type — FFI mirror & projection from the orchestrator wire enum

2.1 The two enums and why both exist

There are **two** TunnelStatus shapes in the system, and they are deliberately not identical:

1. **The orchestrator wire enum** (the broadcast payload, [02-ORCH §4.1]) — the lean five-variant truth the data plane emits: Connecting | Handshaking | Connected{transport, rtt_ms} | Reconnecting | Down{reason}.
2. **The FFI/Dart enum** (the mirror the UI binds, [03 §3.1]) — extends the wire enum with three UI-affordances: Disconnected (clean idle, distinct from an unexpected Down), a path field on Connected ("direct" | "relay"), and Danger{kind} ("leak" | "killswitch_tripped" — drives the red palette).

This document **fixes the projection** between them (§2.3) so the two never drift silently — the projection is the single, auditable place the extension happens, and it is consistent with both [02-ORCH §4.1] (which it consumes verbatim) and [03 §3.1] (which it produces verbatim). The projection lives in the **Rust helix-ffi adapter** (so frb mirrors one enum into Dart — never a hand-written Dart parallel), per [03 §3.2].

2.2 The Rust FFI source (consumed verbatim from [03 §3.1])

```
// helix-ffi/src/api.rs – frb mirrors #[frb(mirror)] types into
// identical Dart types.
// CONSISTENCY: this is the [03 §3.1] enum, byte-for-byte; the
// FFI-contract test
// (§11 FFI-1) asserts the Dart mirror matches it. Do NOT re-
// define it here.
#[frb(mirror)]
pub enum TunnelStatus {
    Disconnected, //
        clean idle (FFI-synth, §2.3)
    Connecting,
    Handshaking,
    Connected { transport: String, path: String, rtt_ms:
        u32 }, // path = "direct" | "relay"
    Reconnecting,
    Down { reason:
        String }, //
        unexpected drop; stable prefix [02-ORCH §4.4]
    Danger { kind:
        String }, // "leak" |
        "killswitch_tripped"
}
```

2.3 The projection adapter (wire → FFI), the single mapping authority

```
// helix-ffi/src/status_adapter.rs
// Subscribes to the orchestrator broadcast [02-ORCH §2.1
//   Orchestrator::subscribe()],
// projects each lean wire event into the extended FFI enum, and
//   re-emits on the
// StreamSink the Dart side listens to (§3). This is the ONLY
//   place the extension
// from 5→7 variants happens. It is a pure function of (wire
//   event, danger signal).
fn project(wire: WireStatus, danger: Option<DangerKind>) ->
  TunnelStatus {
  match (wire, danger) {
    // Danger overrides everything (H-I2): a leak / tripped
    // kill-switch paints red
    // regardless of the underlying wire state.
    (_, Some(DangerKind::Leak)) =>
      TunnelStatus::Danger { kind: "leak".into() },
    (_, Some(DangerKind::KillSwitchTripped))=>
      TunnelStatus::Danger { kind:
        "killswitch_tripped".into() },

    (WireStatus::Connecting, None) =>
      TunnelStatus::Connecting,
    (WireStatus::Handshaking, None) =>
      TunnelStatus::Handshaking,
    (WireStatus::Connected { transport, rtt_ms }, None) =>
      TunnelStatus::Connected { transport, path:
        path_label(), rtt_ms }, // §2.4 path source
    (WireStatus::Reconnecting, None) =>
      TunnelStatus::Reconnecting,

    // The KEY reconciliation: a CLEAN user-stop
    // (Down{reason:"stopped"}) projects to
    // Disconnected (idle), NOT Down. Every OTHER Down reason
    // stays Down. This mirrors the
    // orchestrator's own Idle/ShuttingDown→Connecting/Down
    // internal projection [02-ORCH §5.1].
    (WireStatus::Down { reason }, None) if reason ==
      "stopped" => TunnelStatus::Disconnected,
    (WireStatus::Down { reason }, None) =>
      TunnelStatus::Down { reason },
  }
}
```

Wire enum [02-ORCH §4.1]	Danger signal	FFI/Dart enum [03 §3.1]
(pre-start / post-clean-stop)	—	Disconnected
Connecting	—	Connecting

Wire enum [02-ORCH §4.1]	Danger signal	FFI/Dart enum [03 §3.1]
Handshaking	—	Handshaking
Connected{transport,rtt_ms}	—	Connected{transport,path,rtt_ms}
Reconnecting	—	Reconnecting
Down{reason="stopped"}	—	Disconnected
Down{reason≠"stopped"}	—	Down{reason}
any	Leak / KillSwitchTripped	Danger{kind}

2.4 path and danger provenance (honest boundary, §11.4.6)

- **path** ("direct"|"relay") is a Phase-2 affordance (direct P2P vs DERP-style relay [SYN §4]). UNVERIFIED: the Phase-0/1 wire enum does **not** carry a path field. Until the orchestrator surfaces NAT-traversal state, the FFI adapter sets path from the active transport posture — "relay" for any gateway-routed carrier, "direct" only when the orchestrator reports a hole-punched P2P session [SYN §4]. The default in Phase 1 is "relay"; the StatusChip renders it but **MUST NOT** claim "direct" without that signal.
- **danger** is a separate orchestrator signal, **not** a wire TunnelStatus variant: a DNS-leak detection or a tripped kill-switch firewall rule [02-ORCH §8] raises a DangerKind the adapter folds in. UNVERIFIED: the exact leak-detection mechanism (continuous resolver probe vs firewall-counter) is a doc-02/orchestrator concern; this doc only consumes the resulting DangerKind.

2.5 The Dart-facing type (generated mirror) + a11y projection (CI5)

```
// helix_core_ffilib/frb_generated.dart - GENERATED from §2.2;
// do NOT hand-edit.
sealed class TunnelStatus { const TunnelStatus(); }
class Disconnected extends TunnelStatus { const Disconnected(); }
class Connecting extends TunnelStatus { const Connecting(); }
class Handshaking extends TunnelStatus { const Handshaking(); }
class Connected extends TunnelStatus { final String transport,
    path; final int rttMs;
    const
    Connected(this.transport, this.path, this.rttMs); }
class Reconnecting extends TunnelStatus { const Reconnecting(); }
class Down extends TunnelStatus { final String reason;
    const Down(this.reason); }
class Danger extends TunnelStatus { final String kind;
    const Danger(this.kind); }
```

```
// helix_domain/lib/models/tunnel_status_x.dart - a11y +
// classification helpers (CI5).
// Color is read in helix_design [03 §7.1]; the SEMANTIC LABEL is
// read here, from the
```

```

// SAME source, so "announced, not just colored" cannot drift
// (CI5).
extension TunnelStatusX on TunnelStatus {
  /// The screen-reader announcement – never color alone (CI5).
  String get semanticLabel => switch (this) {
    Disconnected() => 'Not protected. Disconnected.',
    Connecting()   => 'Connecting...',
    Handshaking()  => 'Establishing secure tunnel...',
    Connected(:final transport, :final path, :final rttMs) =>
      'Protected via $transport, $path path, $rttMs
      milliseconds.',
    Reconnecting() => 'Connection dropped. Reconnecting...',
    Down(:final reason) => 'Disconnected unexpectedly: $reason.',
    Danger(:final kind) => kind == 'leak'
      ? 'WARNING: traffic may be leaking. Not protected.'
      : 'WARNING: kill-switch tripped. Traffic is blocked.',
  };

  bool get isProtected    => this is Connected;           // the
  // ONLY protected state
  bool get isTransitional=> this is Connecting || this is
    Handshaking || this is Reconnecting;
  bool get isDanger       => this is Danger;
}

```

Consistency assertion (FFI-1, §11). The Dart sealed class above MUST be the frb-generated mirror of §2.2, and §2.2 MUST equal [03 §3.1] verbatim, which in turn projects [02-ORCH §4.1] via §2.3. The contract test asserts all three links so a drift at any seam FAILs the build [03 §12].

3. The FFI seam — StreamSink → Dart broadcast Stream

3.1 The Rust side: one StreamSink per subscription

```

// helix-ffi/src/api.rs [03 §3.1]
// frb turns this into a Dart `Stream<TunnelStatus>`. The sink is
// fed by the projection
// adapter (§2.3), which is driven by the orchestrator broadcast
// receiver [02-ORCH §4.6].
pub fn status_stream(sink: StreamSink<TunnelStatus>) {
  let mut rx =
    ORCH.get().expect("started").subscribe();           //
    broadcast::Receiver
  tokio::spawn(async move {
    // emit the current state immediately so a late Dart
    // listener is never blank (§3.3)
    let _ = sink.add(project(current_wire(),
      current_danger()));
    loop {
      match rx.recv().await {

```

```

Ok(wire) => { let _ =
sink.add(project(wire, current_danger())); }
Err(RecvError::Lagged(_)) => { let _ =
sink.add(project(current_wire(), current_danger())); } //
re-read latest (H-I5)
Err(RecvError::Closed) =>
break, // orchestrator gone → close the sink
}
}
// sink dropped here → Dart Stream completes (§3.4)
});
}

```

Key properties: (1) the **first** `sink.add` is the *current* projected state, so a Dart `StreamProvider` that subscribes after a connection is already up renders correctly without a race (§3.3); (2) a broadcast `Lagged` is converted to a re-emit of `latest`, **never** propagated as an error (H-I5) [02-ORCH §4.6]; (3) the spawned task ends when the orchestrator's sender closes, completing the Dart stream cleanly.

3.2 The frb threading & isolate model

`flutter_rust_bridge v2` runs Rust `async` on its own worker (a Rust-side Tokio runtime, not the Dart isolate) and marshals each `sink.add(value)` across the isolate boundary onto the Dart event loop [research-flutter_ffi]. Consequences the state layer **MUST** respect:

Concern	Behaviour	State-layer rule
Threading	events arrive on the Dart platform isolate's event queue (frb serializes)	providers may mutate freely; no extra isolate hop needed
Ordering	frb preserves per-sink add order [research-flutter_ffi]	the reducer (§5.1) may assume monotonic delivery within one sink
Backpressure	the Dart side cannot stall the Rust broadcast (the §3.1 loop drops to latest on <code>Lagged</code>)	a slow widget rebuild never wedges the core (H-I5)
Cancellation	dropping the Dart <code>StreamSubscription</code> drops the sink; the §3.1 task exits on next add failure	<code>ref.dispose</code> MUST cancel the subscription (§3.4, §9.2)
Memory	one spawned Rust task + one channel per <code>status_stream()</code> call	call <code>status_stream()</code> once , fan-out in Dart via a broadcast <code>StreamProvider</code> (§3.3, §9.3)

3.3 The Dart wrapper: one sink, broadcast fan-out

```

// helix_core_ffi/lib/helix_core.dart – the idiomatic wrapper [03
§3.2].
class RealHelixCore implements HelixCore {

```

```

Stream<TunnelStatus>? _statusCache;           // single
        underlying sink, fanned out

@override
Stream<TunnelStatus> statusStream() =>
    // frb StreamSink → a single-subscription Dart
    Stream; .asBroadcastStream()
    // lets N widgets watch via the StreamProvider without N
    Rust sinks (§9.3).
    _statusCache ??=
        HelixCoreFfi.instance.statusStream().asBroadcastStream();

@override
Future<void> start({required String transport, String?
    mapPathOrSession,
        CoreMode mode = CoreMode.client}) =>
    HelixCoreFfi.instance.start(cfg: ClientConfig(
        transport: transport, mapPathOrSession:
        mapPathOrSession ?? '', mode: mode));

@override Future<void> stop()                =>
    HelixCoreFfi.instance.stop();
@override Future<List<ExitOption>> exits()    =>
    HelixCoreFfi.instance.exits();
@override Future<void> setExit(String id, {List<String>?
    multiHopChain}) =>
    HelixCoreFfi.instance.setExit(id: id, multiHopChain:
    multiHopChain);
@override Future<void> setShields(Shields s)   =>
    HelixCoreFfi.instance.setShields(s: s);
@override Future<AdvertiseResult> advertise(List<String> cidrs)
    =>
    HelixCoreFfi.instance.advertise(cidrs: cidrs);
@override Future<void> attachTun(int fd)        =>
    HelixCoreFfi.instance.attachTun(fd: fd);
@override Future<void> detachTun()              =>
    HelixCoreFfi.instance.detachTun();
}

/// Console flavor: no Rust staticlib linked (CI3, H-I3).
/// statusStream is a const idle.
class NoCore implements HelixCore {
    @override Stream<TunnelStatus> statusStream() => const
        Stream.empty();
    @override Future<void> start({required String transport,
        String? mapPathOrSession,
            CoreMode mode = CoreMode.client}) async => throw
        UnsupportedError('Console has no tunnel (CI3)');
    // ...every tunnel method throws UnsupportedError; Console never
    // calls them (§4.4).
    @override Future<void> stop() async {}
    @override Future<List<ExitOption>> exits() async => const [];
    @override Future<void> setExit(String id, {List<String>?
        multiHopChain}) async {}
    @override Future<void> setShields(Shields s) async {}
}

```

```

@override Future<AdvertiseResult> advertise(List<String> cidrs)
    async =>
        throw UnsupportedError('Console cannot advertise');
@override Future<void> attachTun(int fd) async {}
@override Future<void> detachTun() async {}
}

```

3.4 Stream lifecycle & disposal

```

sequenceDiagram
    autonumber
    participant W as Widget (ref.watch)
    participant SP as tunnelStatusProvider (StreamProvider)
    participant DC as RealHelixCore (Dart)
    participant FRB as frb StreamSink (Rust task §3.1)
    participant ORCH as orchestrator broadcast

    W->>SP: first watch → provider initializes
    SP->>DC: statusStream() (cached broadcast)
    DC->>FRB: status_stream(sink) (spawns Rust task ONCE)
    FRB->>ORCH: subscribe()
    FRB-->>SP: add(project(current)) %% immediate current state (§3.1)
    ORCH-->>FRB: wire event
    FRB-->>SP: add(project(event))
    SP-->>W: AsyncData(status) → rebuild
    Note over SP: last watcher gone → autoDispose → ref.onDispose cancels sub
    SP->>FRB: StreamSubscription.cancel() → sink dropped → Rust task exits
    FRB->>ORCH: receiver dropped

```

The StreamProvider is **NOT** autoDispose for Access/Connector (the status must survive screen navigation so a backgrounded connect keeps streaming) — it is explicitly keepAlive and disposed only at app teardown (§9.2). The single Rust sink (§3.3) is therefore long-lived; this is intentional and bounded (one task, one channel; §9.3).

4. The provider graph

4.1 The full graph (Mermaid)

```

flowchart TB
    subgraph cfg ["config / DI (overridden in runHelixApp, §4.4)"]
        FLV["flavorProvider<br/>Provider<HelixFlavor>"]
        CAP["capabilitiesProvider<br/>Provider<Set<Capability>>"]
        CORE["helixCoreProvider<br/>Provider<HelixCore>"]
    end
    TP["tunnelPlatformProvider<br/>"]

```

```

>Provider<TunnelPlatform>;<br/>(per-OS impl |
NoTunnelPlatform)"]
    API["apiClientProvider<br/>Provider<HelixApi>"]
    WS["wsClientProvider<br/>Provider<HelixWsClient>"]
end
subgraph tun["tunnel state (Access/Connector)"]
    TS["tunnelStatusProvider<br/
>StreamProvider<TunnelStatus>; (§5.1)"]
    CC["connectControllerProvider<br/>AsyncNotifierProvider
 (§5.2)"]
    INTENT["connectIntentProvider<br/
>NotifierProvider<bool>; (§8.1)"]
    DISP["displayStatusProvider<br/
>Provider<DisplayStatus>; (§8.2)"]
    SH["shieldsProvider<br/
>AsyncNotifierProvider<Shields>; (§5.4)"]
    EX["exitsProvider<br/
>FutureProvider<List<ExitOption>>; (§5.5)"]
    SEL["selectedExitProvider<br/
>NotifierProvider<ExitOption?>"]
end
subgraph ctl["control-plane state (all flavors)"]
    SESS["sessionProvider<br/
>AsyncNotifierProvider<Session>; (§6.2)"]
    EVT["eventStreamProvider<br/
>StreamProvider<ControlEvent>; (§7.1)"]
    DEV["devicesProvider<br/
>NotifierProvider<DeviceList>; (§7.2)"]
    NET["networksProvider<br/
>NotifierProvider<NetworkList>"]
    POL["policyProvider<br/
>AsyncNotifierProvider<Policy>"]
    AUD["auditFeedProvider<br/
>NotifierProvider<AuditFeed>"]
end

FLV --> CORE & TP
CAP --> CORE & TP
CORE --> TS
TS --> DISP
INTENT --> DISP
TP --> CC
SESS --> CC
CORE --> CC & SH & EX
SESS --> API & WS
API --> SESS & DEV & NET & POL & EX
WS --> EVT
EVT --> DEV & NET & POL & AUD
EX --> SEL --> CC

```

4.2 Provider catalogue (type, lifetime, flavor scope)

Provider	Riverpod kind	Lifetime	Flavors
flavorProvider	Provider<HelixFlavor>	keepAlive (overridden)	all
capabilitiesProvider	Provider<Set<Capability>>	keepAlive (overridden)	all
helixCoreProvider	Provider<HelixCore>	keepAlive (overridden)	Access/ Connector real; Console NoCore
tunnelPlatformProvider	Provider<TunnelPlatform>	keepAlive (overridden)	Access/ Connector
apiClientProvider	Provider<HelixApi>	keepAlive	all
wsClientProvider	Provider<HelixWsClient>	keepAlive	Console (+Access for own events)
tunnelStatusProvider	StreamProvider<TunnelStatus>	keepAlive (§3.4)	Access/ Connector
connectControllerProvider	AsyncNotifierProvider<...,void>	keepAlive	Access/ Connector
connectIntentProvider	NotifierProvider<...,bool>	keepAlive	Access/ Connector
displayStatusProvider	Provider<DisplayStatus>	autoDispose	Access/ Connector
shieldsProvider	AsyncNotifierProvider<...,Shields>	keepAlive	Access
exitsProvider	FutureProvider<List<ExitOption>>	autoDispose	Access
selectedExitProvider	NotifierProvider<...,ExitOption?>	keepAlive	Access
sessionProvider	AsyncNotifierProvider<...,Session>	keepAlive	all
eventStreamProvider	StreamProvider<ControlEvent>	keepAlive	Console (+Access)
devicesProvider/ networksProvider/ policyProvider/ auditFeedProvider	NotifierProvider/ AsyncNotifierProvider	autoDispose per screen	Console

4.3 Generator vs hand-written providers (decision)

Both @riverpod codegen (riverpod_generator) and hand-written providers are viable. **Recommendation:** use @riverpod codegen for the family providers (devicesProvider(tenantId), policyProvider(networkId)) where typed arguments matter, and hand-written StreamProvider/Provider for the small fixed

set (status/intent/core) — keeping the FFI-adjacent providers hand-written makes the stream wiring auditable. Codegen drift is checked in the me los drift gate [03 §11]. (UNVERIFIED: the exact riverpod_generator version is a Phase-1 pin.)

4.4 Flavor wiring — runHelixApp overrides (the only DI seam)

```
// helix_domain/lib/run_helix_app.dart [03 §6]
Future<void> runHelixApp({
  required HelixFlavor flavor,
  required Widget home,
  required Set<Capability> capabilities,
}) async {
  final container = ProviderContainer(overrides: [
    flavorProvider.overrideWithValue(flavor),
    capabilitiesProvider.overrideWithValue(capabilities),

    // Only flavors with Capability.tunnel link a real Rust core
    // (CI3, H-I3):
    helixCoreProvider.overrideWith((ref) =>
      capabilities.contains(Capability.tunnel) ?
      RealHelixCore() : NoCore()),

    // The per-OS shim impl is injected here; Console gets a no-
    // op:
    tunnelPlatformProvider.overrideWith((ref) =>
      capabilities.contains(Capability.tunnel)
      ? TunnelPlatform.forCurrentOs() // §5.3 –
      MethodChannel/EventChannel impl
      : const NoTunnelPlatform()),
  ]);
  runApp(UncontrolledProviderScope(container: container,
    child: HelixApp(home: home, theme:
      helixTheme(flavor)))); // [03 §7]
}
```

The override list is the **single** place flavor capability gates the graph: Console's NoCore/NoTunnelPlatform mean its widget tree never even references tunnelStatusProvider, so the Rust staticlib is tree-shaken (CI3) [03 §6].

4.5 Test override seam (H-I4)

```
// test: same generated model types, faked behaviour (no mocks-
// claiming-PASS, [03 §12]).
ProviderContainer(overrides: [
  helixCoreProvider.overrideWithValue(FakeHelixCore(scriptedStatus: [
    const Connecting(), const Handshaking(), const
    Connected('masque-h3', 'relay', 23),
  ])), // drives
  a real StreamProvider
  tunnelPlatformProvider.overrideWithValue(FakeTunnelPlatform()),
  apiClientProvider.overrideWithValue(FakeHelixApi.fromFixtures()),
]);
```

FakeHelixCore emits the **generated** TunnelStatus types (never a parallel stub), so a state test exercises the real contract — the §11 STATE/FFI invariant.

5. tunnelStatusProvider + ConnectController (Access/Connector)

5.1 The status provider (the truth binding)

```
// helix_domain/lib/providers/tunnel.dart
final tunnelStatusProvider = StreamProvider<TunnelStatus>((ref) {
  final core = ref.watch(helixCoreProvider);
  ref.keepAlive(); // survive
  navigation (§3.4)
  return core.statusStream(); // frb
  StreamSink → broadcast Stream (CI2)
});
```

AsyncValue<TunnelStatus> semantics the UI relies on:

AsyncValue	Meaning at the UI	Rendered state
AsyncLoading (no value yet)	provider initialized, first sink.add not arrived	treat as Disconnected placeholder (§3.1 emits current immediately, so this is sub-frame)
AsyncData(s)	latest projected status	render s
AsyncError(e)	the Dart stream errored (should be unreachable — §3.1 never adds errors)	log + render last good; surface as Down{reason:"ffi-stream- error"} (§10)

5.2 ConnectController — commands only (never writes rendered state)

```
// helix_domain/lib/providers/connect_controller.dart
final connectControllerProvider =
  AsyncNotifierProvider<ConnectController,
    void>(ConnectController.new);

class ConnectController extends AsyncNotifier<void> {
  @override Future<void> build() async {} // no state
  of its own (CI2)

  /// Issue the CONNECT command. Does NOT set "connected" – the
  stream does (§1).
  Future<void> connect({String transport = 'auto'}) async {
    state = const AsyncLoading();
    ref.read(connectIntentProvider.notifier).set(true); //
    optimistic intent (§8.1)
    state = await AsyncValue.guard(() async {
      final cfg = await ref.read(sessionProvider.future)
```

```

        .then((s) =>
s.resolveTunnelConfig( // §6.2 → TunnelConfig [03 §4]
        exit:
ref.read(selectedExitProvider)));
await
ref.read(tunnelPlatformProvider).startTunnel(cfg); //
shim creates TUN + links core
// NOTE: returns when the COMMAND is accepted, NOT when
Connected. The
// tunnelStatusProvider stream is the sole source of the
green state (CI2).
});
if (state.hasError) {

    ref.read(connectIntentProvider.notifier).set(false); //
    command failed → drop intent
}
}

Future<void> disconnect() async {
    ref.read(connectIntentProvider.notifier).set(false);
    state = await AsyncValue.guard(
        () => ref.read(tunnelPlatformProvider).stopTunnel());
}

/// Toggle off the LATEST stream value (the actual state), not
internal guesswork.
Future<void> toggle(TunnelStatus? s) =>
    (s is Connected) ? disconnect() : connect();
}

// the ConnectButton – a pure function of the stream (CI2),
command on tap (§1)
final status = ref.watch(tunnelStatusProvider);
final display = ref.watch(displayStatusProvider); // intent-vs-
actual blend (§8.2)
ConnectButton(
    state: status.valueOrNull ?? const Disconnected(),
    display: display, // shows
        "Reconnecting..." honestly (§8)
    busy: ref.watch(connectControllerProvider).isLoading,
    onTap: () =>
        ref.read(connectControllerProvider.notifier).toggle(status.valueOrNull),
);

```

5.3 TunnelPlatform injection (the command channel)

```

// helix_core_ffl/lib/tunnel_platform.dart [03 §4] – the abstract
contract (recap).
abstract class TunnelPlatform {
    Future<void> startTunnel(TunnelConfig cfg);
    Future<void> stopTunnel();
    Stream<PlatformTunnelEvent> events(); // up/down/
        permissionDenied/revoked/error
}

```

```

/// Factory selecting the per-OS MethodChannel/EventChannel impl
 (§4.4).
static TunnelPlatform forCurrentOs() => switch
  (defaultTargetPlatform) {
    TargetPlatform.iOS || TargetPlatform.macOS => const
      _AppleTunnelPlatform(),
    TargetPlatform.android                      => const
      _AndroidTunnelPlatform(),
    TargetPlatform.windows                     => const
      _WindowsPipeTunnelPlatform(),
    TargetPlatform.linux                      => const
      _LinuxTunnelPlatform(),
    -                                         => const
      _ChannelTunnelPlatform(), // ohos/aurora
  };
}

```

The `events()` stream (lifecycle: `permissionDenied/revoked/error`) is a **separate** signal from `tunnelStatusProvider` (protection state). The state layer folds it: a `revoked` event (OS or admin killed the tunnel out-of-band, [03 §4.1 O3]) sets `intent=false` and lets the core stream's subsequent `Down/Danger` paint the truth — the UI never claims connected after a revoke [03 §4.1 O3, CI2].

```

// helix_domain/lib/providers/platform_events.dart
final platformEventsProvider =
  StreamProvider<PlatformTunnelEvent>((ref) {
    ref.keepAlive();
    return ref.watch(tunnelPlatformProvider).events();
  });

// folded by a listener wired once at app start (§9.1):
void wirePlatformEvents(Ref ref) {
  ref.listen(platformEventsProvider, (_, next) {
    final ev = next.valueOrNull; if (ev == null) return;
    switch (ev.kind) {
      case PlatformTunnelEventKind.permissionDenied:

        ref.read(connectIntentProvider.notifier).set(false); //
        honest: not connecting
      case PlatformTunnelEventKind.revoked:
      case PlatformTunnelEventKind.error:

        ref.read(connectIntentProvider.notifier).set(false); //
        core stream paints Down/Danger
      case PlatformTunnelEventKind.up:
      case PlatformTunnelEventKind.down:

        break; //
        the core stream is authoritative
    }
  });
}

```

5.4 shieldsProvider — logic-only, no OS tunnel (CI1)

```
// helix_domain/lib/providers/shields.dart – setShields is pure
  core logic (no shim) [03 §3.3].
final shieldsProvider = AsyncNotifierProvider<ShieldsController,
  Shields>(ShieldsController.new);

class ShieldsController extends AsyncNotifier<Shields> {
  @override Future<Shields> build() async =>
    const Shields(killSwitch: true, dnsProtection: true, daita:
      false,
                postQuantum: false, splitTunnel: []);
  Future<void> update(Shields s) async {
    state = const AsyncLoading();
    state = await AsyncValue.guard(() async {
      await
        ref.read(helixCoreProvider).setShields(s);           //
        UI->FFI direct (CI1, no shim)
      return s;
    });
  }
}
```

5.5 exitsProvider + selectedExitProvider

```
final exitsProvider =
  FutureProvider.autoDispose<List<ExitOption>>((ref) async {
    final list = await
      ref.watch(helixCoreProvider).exits();           //
      core.exits() [03 §3]
    list.sort((a, b) => (a.rttMs ?? 1 << 30).compareTo(b.rttMs ?? 1
      << 30)); // RTT-sorted [03 §7.2]
    return list;
  });
final selectedExitProvider = NotifierProvider<SelectedExit,
  ExitOption?>(SelectedExit.new);
class SelectedExit extends Notifier<ExitOption?> {
  @override ExitOption? build() => null;
  Future<void> choose(ExitOption e, {List<String>? chain}) async {
    await ref.read(helixCoreProvider).setExit(e.id,
      multiHopChain: chain); // logic-only (CI1)
    state = e;
  }
}
```

6. The helix_api client — REST + WS/SSE

6.1 Package surface

helix_api is a decoupled, reusable package [03 §2]: a **generated** OpenAPI REST client (rest_generated.dart) plus a hand-written WS/SSE client (ws_client.dart) [04_UI §1]. The wire schema, auth, and event payloads are doc 02's; this doc fixes only the Dart consumption surface.

```
// helix_api/lib/helix_api.dart – the typed facade over
// rest_generated.dart (doc 02 schema).
abstract class HelixApi {
  // auth / enrollment (doc 02):
  Future<Session> enrollOidc(String idToken);
  Future<Session> enrollDeviceToken(String enrollToken,
    WgPublicKey devicePub);
  Future<Session> refresh(Session expiring);
  // resources (Console + Access):
  Future<List<Device>> listDevices(TenantId t);
  Future<void> revokeDevice(DeviceId d); // →
    device.revoked event (doc 02)
  Future<List<Network>> listNetworks(TenantId t);
  Future<Policy> getPolicy(NetworkId n);
  Future<PolicyDiff> previewPolicy(NetworkId n, PolicyDraft
    draft); // effect-diff [03 §7.2]
  Future<void> applyPolicy(NetworkId n, PolicyDraft
    draft);
  Future<EnrollToken> mintEnrollToken(NetworkId n); // QR
    enroll [03 §13 T1.6]
}
```

6.2 sessionProvider — the auth root every repo depends on

```
final sessionProvider = AsyncNotifierProvider<SessionController,
  Session>(SessionController.new);

class SessionController extends AsyncNotifier<Session> {
  @override Future<Session> build() async {
    final stored = await
      ref.read(secureStoreProvider).readSession(); //
      §11.4.10: secure storage
    if (stored == null) throw const UnauthenticatedException();
    if (stored.isExpiring) return
      ref.read(apiClientProvider).refresh(stored);
    // schedule a refresh before expiry so REST/WS never 401 mid-
    // session:
    ref.onDispose(_refreshTimer.cancel);
    _scheduleRefresh(stored);
    return stored;
  }
  // resolveTunnelConfig: turn the session + selected exit into
  // the [03 §4] TunnelConfig the shim needs.
}
```

apiClientProvider and wsClientProvider ref.watch(sessionProvider) so a token refresh transparently re-bears every request and re-auths the socket (§6.3). The **device WG private key never leaves the device** [SYN §7] — only its public key + the session token cross helix_api (the keypair is shim/secure-store owned, doc 06).

6.3 The WS/SSE client (HelixWsClient)

```
// helix_api/lib/ws_client.dart – live control-plane events (doc
    02 GET /v1/stream).
class HelixWsClient {
  HelixWsClient(this._session);
  final Session _session;

  /// Long-lived event stream. Auto-reconnects with backoff;
  /// replays from the last
  /// acked cursor so no event is lost across a flap (doc 02
  /// stream resume semantics).
  Stream<ControlEvent> events() async* {
    var backoff = const Duration(milliseconds: 500);
    var cursor = await _store.lastCursor();
    while (true) {
      try {
        final socket = await _connect(_session.bearer, since:
          cursor); // WS upgrade or SSE fallback
        backoff = const Duration(milliseconds:
          500); // reset on success
        await for (final raw in socket) {
          final ev =
            ControlEvent.fromJson(raw); // doc
            02 schema
          cursor = ev.cursor; await
            _store.saveCursor(cursor); // at-least-once
          yield ev;
        }
      } on SocketException {
        await Future.delayed(backoff);
        backoff = _capped(backoff * 2, const Duration(seconds:
          30)); // jittered, capped
      }
    }
  }
}
```

Transport choice (§11.4.6). UNVERIFIED: whether the control stream is WS or SSE per platform — doc 02 fixes the wire (GET /v1/stream [04_UI §4.3]). The Dart client **MUST** support **both** (WS where bi-directional acks are wanted, SSE where a plain EventSource suffices — Web/Console) and present the **same** Stream<ControlEvent> upward so the reducer (§7) is transport-agnostic.

7. Console: folding control-plane events into Riverpod state

7.1 The event provider + taxonomy

```
// helix_domain/lib/providers/events.dart
final eventStreamProvider = StreamProvider<ControlEvent>((ref) {
  ref.keepAlive();
  return ref.watch(wsClientProvider).events(); // §6.3
});
```

The closed event taxonomy the reducer handles [04_UI §4.3, doc 02] — Console folds each into the relevant collection so the device list, topology view, and audit feed update **live without refresh** [03 §8.3]:

Event	Payload (doc 02)	Reduced into
device.online / device.offline	{deviceId, at}	devicesProvider presence
device.revoked	{deviceId, reason}	devicesProvider (mark revoked) + (Access self) drop intent (§5.3)
route.changed	{networkId, delta}	networksProvider topology
policy.compiled	{networkId, version, hash}	policyProvider version + audit
enroll.completed	{deviceId, networkId}	devicesProvider add
audit.appended	{actor, action, at}	auditFeedProvider prepend

7.2 The reducer (fold events into collection state)

```
flowchart TB
    WS["wsClientProvider.events()"] -->
    EV["eventStreamProvider<br/>StreamProvider<ControlEvent>"]
    EV -->|ref.listen| RED["per-collection reducers (§7.2)"]
    subgraph RED
        D["devicesProvider.apply(ev)"]
        N["networksProvider.apply(ev)"]
        P["policyProvider.apply(ev)"]
        A["auditFeedProvider.prepend(ev)"]
    end
    end
    SNAP["REST snapshot on screen open<br/>(devicesProvider build  
→ api.listDevices)"] --> D
    D --> UI1["DeviceList (live)"]
    N --> UI2["TopologyView (live)"]
    A --> UI3["AuditFeed (live)"]
```

```

// helix_domain/lib/providers/devices.dart – snapshot from REST,
// then live deltas from WS.
final devicesProvider = NotifierProvider<DevicesController,
    DeviceList>(DevicesController.new);

class DevicesController extends Notifier<DeviceList> {
  @override DeviceList build() {
    final tenant = ref.watch(currentTenantProvider);
    // initial snapshot (REST) – render immediately, then live-
    // patch from events:
    ref.read(apiClientProvider).listDevices(tenant).then((d) =>
        state = DeviceList(d));
    // subscribe to the live stream; fold each relevant event:
    ref.listen(eventStreamProvider, (_, next) {
      final ev = next.valueOrNull; if (ev == null) return;
      state = switch (ev) {
        DeviceOnline(:final id) => state.markOnline(id),
        DeviceOffline(:final id) => state.markOffline(id),
        DeviceRevoked(:final id) => state.markRevoked(id),
        EnrollCompleted(:final device) => state.add(device),
        _ => state, // not
        ours → ignore
      };
    });
    return const DeviceList.empty();
  }
}

```

The pattern is **snapshot-then-stream**: a REST list* for the initial state (so the screen is not blank on open), then eventStreamProvider deltas patch it in place — the same "snapshot+deltas, push-don't-poll" shape the coordinator uses on the data plane [SYN §2, 04_UI §4.3]. The reducer is a pure (state, event) → state function (testable per §11 STATE).

7.3 Idempotency & ordering (no double-apply)

- Each ControlEvent carries a monotonic cursor (§6.3). The reducer is **idempotent**: re-applying an event already folded (post-reconnect replay) is a no-op (e.g. markOnline of an already-online device). This is required because the WS resume (§6.3) is **at-least-once**, not exactly-once.
 - A snapshot that arrives *after* some deltas (slow REST, fast WS) reconciles by **last-writer-by-cursor**: deltas with a cursor newer than the snapshot's as_of are re-applied on top of the snapshot. UNVERIFIED: the exact snapshot as_of field is doc 02's; the reducer assumes it exists and SKIPS reconciliation with an honest log line if absent (§11.4.6).
-

8. Offline / optimistic — the intent-vs-actual honesty machine

8.1 connectIntentProvider — local wish, never truth

```
// helix_domain/lib/providers/intent.dart
final connectIntentProvider = NotifierProvider<ConnectIntent,
    bool>(ConnectIntent.new);
class ConnectIntent extends Notifier<bool> {
  @override bool build() =>
    false; // default: user wants
    disconnected
  void set(bool wantConnected) => state = wantConnected;
}
```

Intent is set true on connect() and false on disconnect()/revoked/command failure (§5.2/§5.3). It is **never** read as protection state — it only refines *how* a transitional/idle status is *labelled* (§8.2). This is the literal "distinguish intended from actual" rule [04_UI §4.4, 03 §8.4].

8.2 displayStatusProvider — blend (actual is authoritative)

```
// helix_domain/lib/providers/display_status.dart
enum DisplayState { disconnected, connecting, reconnecting,
    connected, danger }
class DisplayStatus {
  final DisplayState state; final TunnelStatus actual; final bool
    intended;
  const DisplayStatus(this.state, this.actual, this.intended);
}

final displayStatusProvider =
  Provider.autoDispose<DisplayStatus>((ref) {
    final actual = ref.watch(tunnelStatusProvider).valueOrNull ??
      const Disconnected();
    final intended = ref.watch(connectIntentProvider);

    // RULE (H-I1, H-I2): ACTUAL is authoritative for color/
    // protection. Intent only
    // refines the LABEL of a non-connected state so the UI never
    // freezes on a stale
    // "Disconnected" while a connect is genuinely in flight.
    final state = switch (actual) {
      Danger() =>
        DisplayState.danger, // overrides
        intent (H-I2)
      Connected() =>
        DisplayState.connected, // actual
        green only
      Connecting() || Handshaking() => DisplayState.connecting,
      Reconnecting() => DisplayState.reconnecting,
      // idle but the user WANTS connected (e.g. control channel
      // flaky, command sent,
```

```

    // stream not yet flowing) → show "connecting...", NOT a
    // misleading green, NOT a
    // dead "disconnected" (honest in-between):
    Disconnected() || Down() => intended ?
        DisplayState.connecting : DisplayState.disconnected,
};
return DisplayStatus(state, actual, intended);
});

```

actual (stream)	intended	DisplayState	Color [03 §7.1]
Connected	any	connected	green stateConnected
Danger	any	danger	red stateDanger (override)
Connecting/ Handshaking	any	connecting	amber stateConnecting
Reconnecting	any	reconnecting	amber stateConnecting
Disconnected/Down	true	connecting	amber (honest "trying")
Disconnected/Down	false	disconnected	grey stateDisconnected

The honesty guarantee (H-I1): the only path to connected (green) is an actual Connected event — intent can never produce green. The only thing intent does is upgrade a *grey idle* to an *amber connecting* while a command is in flight, so the UI is neither lying-green nor frozen-grey.

8.3 Danger is non-suppressible (H-I2)

A `Danger{leak|killswitch_tripped}` projection (§2.3) maps to `DisplayState.danger` **before** any intent check, paints red immediately, and cannot be optimistically hidden. This mirrors the orchestrator's fail-closed posture [02-ORCH §8.4]: the data plane has closed the kill-switch / detected a leak, and the UI must say so the instant the event arrives — the user's "I want connected" intent is irrelevant to a safety alarm.

8.4 Offline control channel ≠ offline tunnel

A flaky **control** channel (REST/WS to the coordinator, §6) is independent of the **data** tunnel (the core stream, §5): the tunnel keeps forwarding on its last-applied map even if the coordinator is unreachable (fail-static [02-ORCH §6.5, I3]). The state layer reflects this honestly:

```

stateDiagram-v2
    [*] --> Idle
    Idle --> Connecting : user connect() → intent=true
    Connecting --> Connected : core stream Connected (ACTUAL, not
intent)
    Connecting --> Connecting : control WS down (irrelevant – data
path is core stream)
    Connected --> Reconnecting : core stream Reconnecting (data
drop)

```

```

    Connected --> Danger : core stream Danger (leak/killswitch) –
overrides intent
    Reconnecting --> Connected : core stream Connected
    Connected --> Idle : user disconnect() → intent=false → stream
Disconnected
    Danger --> Idle : user disconnect()
    note right of Connected
        A control-plane (REST/WS) outage NEVER changes this machine
–
    only the CORE STATUS STREAM does (CI2, fail-static).
end note

```

A WS reconnect banner ("reconnecting to console...") is a **separate** UI affordance driven by eventStreamProvider's AsyncError/reconnect state — it MUST NOT be confused with the tunnel's Reconnecting (different stream, different meaning).

9. App lifecycle, provider disposal & memory budgets

9.1 Startup wiring (once, at runHelixApp)

```

// after runApp: wire the cross-cutting listeners that fold side-
streams (§5.3, §7.2).
// Done via a top-level Consumer/ref so they live for the app's
lifetime and are
// disposed only at teardown.
class HelixApp extends ConsumerStatefulWidget { /* ... */ }
class _HelixAppState extends ConsumerState<HelixApp> with
    WidgetsBindingObserver {
    @override void initState() {
        super.initState();
        WidgetsBinding.instance.addObserver(this);
        if (ref.read(capabilitiesProvider).contains(Capability.tunnel)) {
            wirePlatformEvents(ref); // §5.3 –
            fold shim lifecycle events
        }
    }
    @override void dispose() {
        WidgetsBinding.instance.removeObserver(this);
        super.dispose(); }

    @override void didChangeAppLifecycleState(AppLifecycleState s) {
        // §9.4: on resume, force-refresh the control snapshot (WS may
        have flapped while backgrounded);
        // the tunnel status stream needs no action (the core kept
        running in its OS process, [03 §4]).
        if (s == AppLifecycleState.resumed)
            ref.invalidate(devicesProvider);
    }
}

```

9.2 Disposal rules

Provider class	Disposal	Why
tunnelStatusProvider, platformEventsProvider, eventStreamProvider	keepAlive	streams must survive screen navigation; one Rust sink / one WS (§3.4)
helixCoreProvider, tunnelPlatformProvider, apiClientProvider, wsClientProvider, sessionProvider	keepAlive	app-singletons; disposed only at container teardown
exitsProvider, screen devicesProvider/policyProvider	autoDispose	re-fetched on screen open; freed when the screen pops
connectIntentProvider, selectedExitProvider, shieldsProvider	keepAlive	user state spanning screens

Every provider that holds a `StreamSubscription` **MUST** cancel it in `ref.onDispose` (`StreamProvider` does this automatically; hand-folded `ref.listen` subscriptions are auto-cancelled when the host provider disposes). A leaked subscription is a §11 MEM defect.

9.3 Memory budget (state layer)

Item	Budget	How
Rust FFI status sinks	exactly 1 live <code>status_stream()</code> task	Dart caches + <code>asBroadcastStream()</code> (§3.3) — N widgets, 1 sink
WS/SSE socket	exactly 1 per session	<code>wsClientProvider</code> singleton (§6.3)
Event history retained	bounded ring (audit feed cap, e.g. last 500)	<code>auditFeedProvider</code> caps + drops oldest (no unbounded growth)
Provider graph footprint	low-tens of KB	sealed states, no large caches; heavy Console tables are <code>autoDispose</code> + deferred-loaded [03 §7.3]
Rebuild cost	only widgets watching the changed provider rebuild	granular providers (<code>status</code> $\neq$ <code>devices</code> $\neq$ <code>policy</code>); select for field-level watches

These sit under the app-wide budgets [03 §10]: idle memory single-digit→low-tens MB; the *tunnel core* memory is the Rust budget (the iOS NE ceiling, [03 §5.1]), **not** the Dart state layer's.

9.4 Background / resume

- **Access/Connector:** the tunnel runs in a separate OS process/extension [03 §0.1 C11], so backgrounding the Flutter app does **not** stop the tunnel; on resume,

tunnelStatusProvider re-reads the current state via the §3.1 immediate-emit, so the UI is correct within a frame — no re-poll.

- **Console:** on resume, the WS may have dropped while backgrounded; eventStreamProvider's auto-reconnect (§6.3) replays from the cursor, and didChangeAppLifecycleState invalidates the visible collection so a fresh REST snapshot reconciles any missed deltas (§7.3).

10. Error handling & edge cases

#	Edge case	Required state-layer behaviour	Source
C1	FFI stream emits nothing yet (cold start)	§3.1 emits <i>current</i> immediately; AsyncLoading is sub-frame; render Disconnected placeholder	§3.1, §5.1
C2	broadcast Lagged (UI wedged > 64 events behind)	§3.1 converts to a re-emit of latest; provider never sees an error (H-I5)	[02-ORCH §4.6]
C3	connect() command throws (shim permissionDenied)	ConnectController sets AsyncError, intent←false; displayStatus→disconnected; show permission rationale (NOT error red — distinct from Danger) [03 §4.1 O1]	§5.2, §5.3
C4	OS revoked event mid-Connected	platform-events listener sets intent←false; core stream emits Down/Danger; UI paints honestly, never stays green [03 §4.1 O3]	§5.3
C5	control WS down while tunnel Connected	tunnel UI unaffected (different stream, fail-static); only a "reconnecting to console" banner shows	§8.4
C6	session token expired mid-use	sessionProvider auto-refreshes (§6.2); REST/WS re-bear; on hard-fail → UnauthenticatedException routes to re-enroll	§6.2
C7	duplicate control event after WS replay	idempotent reducer no-ops (§7.3)	§7.3
C8	snapshot arrives after deltas	last-writer-by-cursor reconciliation; SKIP-with-log if as_of absent (§11.4.6)	§7.3
C9	Danger arrives while user intent=true	displayStatus→danger (red) regardless of intent (H-I2)	§8.3
C10	Console screen watches tunnelStatusProvider (mis-wire)	NoCore().statusStream() = Stream.empty() → AsyncLoading forever; a lint/test (§11 STATE) flags any Console widget referencing tunnel providers (H-I3)	§3.3, §4.4

#	Edge case	Required state-layer behaviour	Source
C11	rapid toggle (connect→disconnect→connect)	AsyncNotifier.guard serializes; toggle reads the <i>latest</i> stream value, not stale intent; no torn state	§5.2
C12	FFI stream errors (should be unreachable)	tunnelStatusProvider surfaces AsyncError; mapped to synthetic Down{reason:"ffi-stream-error"}; logged; NOT shown as Danger (it is an app fault, not a leak)	§5.1

No-guessing on causes (§11.4.6). Error labels are observational, never guessed: permissionDenied (C3) comes from the shim's explicit PlatformTunnelEventKind.permissionDenied [03 §4], never inferred from a generic failure; an ambiguous failure is Down{reason} with the shim's honest detail string, never silently upgraded to Danger.

11. Test points — tied to the §11.4.169 closed test-type vocabulary

Per §11.4.169 every workable item declares its required test types from the CLOSED set (unit · integration · e2e · full-automation · Challenges · HelixQA · DDoS/load · security · stress+chaos · concurrency · race · memory · benchmarking); the ONLY permitted absence of a warranted type is an honest §11.4.3 SKIP-with-reason, never a silent gap; every PASS cites rock-solid captured evidence (§11.4.5/.69/.107). Four-layer §11.4.4(b) enforcement applies to every closure; each meta-test ships a paired §1.1 mutation.

Test id	§11.4.169 type	What it proves for the state layer	Rig / captured evidence
ST-1	unit	project() (§2.3) maps every (wire, danger) pair → exact FFI enum; Down{stopped} → Disconnected, danger overrides	table-driven test over matrix
ST-2	unit	displayStatusProvider (§8.2) never yields connected without actual==Connected; Danger overrides intent	truth-table over (actual intent)
ST-3	unit	the §7.2 reducer is a pure idempotent (state,event)→state; double-apply = no-op	golden event sequence replay
ST-4	integration (real System, infra via containers submodule §11.4.76)	tunnelStatusProvider reflects a REAL helix-core stream over frb on Linux; helix_api hits a booted coordinator	netns + booted gateway coordinator; provider-capture log
ST-5	e2e	full Access journey: tap Connect → Connecting→Handshaking→Connected{masquerade}	Flutter-Linux app; wi-scoped MP4 + vision (§11.4.159) of the State

Test id	§11.4.169 type	What it proves for the state layer	Rig / captured evidence
		h3, relay, 23} rendered ONLY from the stream; tap → Disconnected	
ST-6	full-automation (deterministic §11.4.50)	N=3 identical runs of the connect→status arc produce byte-identical rendered-state traces	scripted FakeHelixC identical trace artifacts
ST-7	Challenges (challenges submodule §11.4.27(B))	a Challenge bank scores PASS only on the captured ST-5 status-arc MP4 + the no-green-on-intent assertion, not config	Challenge result.js
ST-8	HelixQA (helix_qa submodule)	autonomous QA session drives connect/disconnect/revoke across flavors; asserts honest display state each step	HelixQA session evidence bundle
ST-9	DDoS / load-flood	Console WS event-flood (≥10k events/s) folds without dropping UI frames or unbounded growth; audit ring caps	event-flood harness; fold time + RSS CSV
ST-10	security (§11.4.10 + security submodule)	no Session bearer / WG key / SecretBytes ever logged or rendered; secure-store read/write only	plant-and-grep over logs; tracked files (grep-emitted proof)
ST-11	stress + chaos (§11.4.85)	control WS killed mid-Connected → tunnel UI unchanged (fail-static); revoke mid-connect → honest Down; rapid toggle (C11) no torn state	chaos harness: WS SH revoke injection, 100% recovery trace
ST-12	concurrency / atomicity	concurrent stream event + command + WS event mutating providers leaves consistent state; guard serializes commands	concurrency harness on connectController
ST-13	race / deadlock	provider re-entrancy / ref.listen fold during dispose has no use-after-dispose / deadlock	Riverpod ProviderC stress + dispose-during
ST-14	memory	exactly 1 live frb status sink + 1 WS socket regardless of widget count; no subscription leak across 1000 navigations	sink/socket counter + a navigation soak
ST-15	benchmarking / performance	only providers whose value changed trigger rebuilds; status emit → frame within budget; no jank on the connect transition	rebuild-count + frame CSV (select granularity)
FFI-1	unit (contract)	the Dart TunnelStatus mirror == §2.2 Rust == [03 §3.1] == projection of [02-ORCH §4.1]; a drift FAILs the build	shared generated fixtures for-byte enum diff [03

Anti-bluff coupling (§11.4.107/§11.4.158). A green widget/state unit test is NOT proof the user sees the truth. ST-5 (e2e, window-scoped MP4 + vision verdict of the StatusChip reading MASQUE · relay · 23ms) + ST-4 (real-core integration) are the load-bearing evidence; the unit layer (ST-1..3, FFI-1)

exercises real contracts via the *generated* model types (H-I4), never mocks-claiming-PASS [03 §12]. Honest §11.4.3 SKIP applies where a type is not warranted at this layer (e.g. ST-9 DDoS for Access — a pure end-user app has no server-side flood surface; the Console event-flood IS the warranted instance, so it is NOT skipped).

12. Frozen contracts this document fixes

Contract	Fixed value	Consumed by
wire→FFI TunnelStatus projection (§2.3) — the single 5→7 variant mapping authority	§2.3 table	the FFI adapter [03 §3], FFI-1 test
tunnelStatusProvider = the sole rendered-state source (CI2)	§5.1	every reactive Access/Connector widget
command/truth asymmetry — controllers issue commands, never write rendered state	§1, §5.2	ConnectController, ST-5
intent-vs-actual blend (displayStatusProvider) — green only on actual Connected; Danger overrides	§8.2	ConnectButton, the honesty guarantee
snapshot-then-stream reducer (idempotent, cursor-ordered)	§7.2/ §7.3	Console live collections
provider override DI seam (runHelixApp) — flavor + test wiring	§4.4/ §4.5	the three flavors; all state tests
single frb sink + single WS socket memory invariant	§3.3/ §9.3	ST-14 MEM gate

Sources verified

- `final/03-client-core-and-ui.md` [03] — §0.1 (client invariants CI1–CI6), §3 (FFI surface, TunnelStatus mirror + extension note, FFI⇌shim seam), §4 (TunnelPlatform contract + obligations), §6 (flavor/capability gating, runHelixApp), §7 (tokens/state→color, ally CI5), §8 (Riverpod data layer, tunnel-state provider, Console live data, offline/optimistic), §12 (testing + FFI contract test).
- `final/v02-data-plane/orchestrator-and-state.md` [02-ORCH] — §4.1 (the lean wire TunnelStatus enum), §4.4 (Down.reason prefixes), §4.6 (receiver contract / Lagged), §5.1 (OrchState internal-vs-projection), §6.5 (fail-static), §8 (kill-switch/DNS / Danger source), §13.1 (single state authority O-I10), §14 (test points / §11.4.169).
- `final/v02-data-plane/transport-trait.md` [02-TT] — §2 (Transport trait), §5.2 (tunnel-level TunnelStatus projection note), §9 (§11.4.169 test-type table precedent).
- `04_VPN_CLD/HelixVPN-helix-ui-Flutter.md` [04_UI] — §1 (melos packages, Console-no-core-ffi), §4 (Riverpod choice, layering, tunnelStatusProvider, ConnectController, Console live data §4.3,

offline/optimistic §4.4), §5 (FFI surface HelixCore), §6 (TunnelPlatform MethodChannel+EventChannel), §7 (responsive), §12 (state tests with fake core + fake API).

- 04_VPN_CLD/HelixVPN-Architecture-Refined.md [04_ARCH] — §5 (two-cores strategy, FFI via flutter_rust_bridge v2 + UniFFI, why Flutter), §5.6/§5.7 (iOS NE ceiling, Console-web/CI3).
- v09-research/_SYNTHESIS.md [SYN] — §2 (push-don't-poll, event-driven map), §4 (phases — direct-P2P/relay path is Phase 2), §5 (client architecture, Riverpod = pure function of core status stream), §7 (no-logging, keys never leave device), §9 (constitution bindings).
- [research-flutter_ffl] / [research-ios_android] — frb v2 StreamSink→Dart Stream isolate/threading model; per-OS shim channel mechanics. UNVERIFIED items (path-field provenance §2.4, WS-vs-SSE per platform §6.3, exact riverpod_generator pin §4.3) flagged inline per §11.4.6, never fabricated.

Constitution: §11.4.44 (revision header), §11.4.6 (no-guessing — UNVERIFIED markers, observational error labels), §11.4.66 (decisions surfaced not silently resolved), §11.4.5/§11.4.69/§11.4.107/§11.4.158/§11.4.159 (captured/window-scoped evidence), §11.4.169 (comprehensive test-type coverage), §11.4.10 (no secret leak — ST-10), §11.4.28/§11.4.74 (decoupled reusable packages), §11.4.50 (deterministic FA), §11.4.85 (stress+chaos), §11.4.27 (Challenges/HelixQA), §1.1 (paired mutations).

End of nano-detail specification — Volume 4 (Clients), state-management.md. Pairs with 03-client-core-and-ui.md (the client overview this deepens), v02-data-plane/orchestrator-and-state.md (the TunnelStatus stream + state machine this consumes), and v02-data-plane/transport-trait.md (the carrier the Connected.transport label names). The wire→FFI projection (§2.3), the provider graph (§4), and the intent-vs-actual honesty machine (§8) are the frozen state-layer contracts; their implementations may evolve, their shapes may not.